

Last class :

Reference Model

↳ TCP

↳ OSI/ISO

Layers

↳ 5-7

Application layer

↳ data unit

↳ message

Transport layer

↳ data unit
segment

Through put

Latency and RTT

bandwidth 25Mbps

throughput

↳ max successful
outcome in IRL → 15/20 Mbps

*** # E2E Throughput and Delay

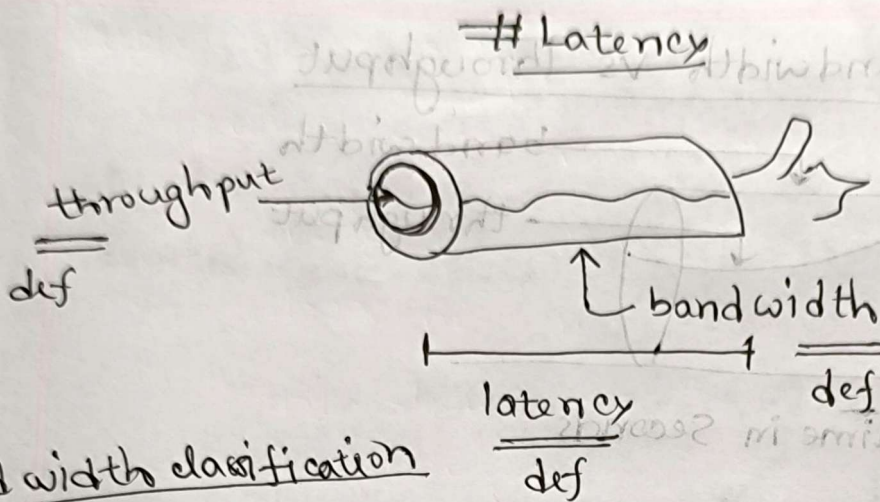
< This diagram is important >

E2E delay → { sum of links
in the optimal path }

Throughput

↳ bottleneck

data transmission
rate



Band width classification

- ① Access ISP 25Mbps
↓ ↑
- ② Tar 2 ISP 1 Gbps
↓ ↑
- ③ Tar 1 ISP 5 Gbps

Latency → delay from source to destination

RTT → round trip time

Throughput, Delay, Loss

Router → has buffer ↑ delay

↓ buffer '0'

* Queuing Delay > Total

Latency ↔
RTT ↔

Throughput Classification

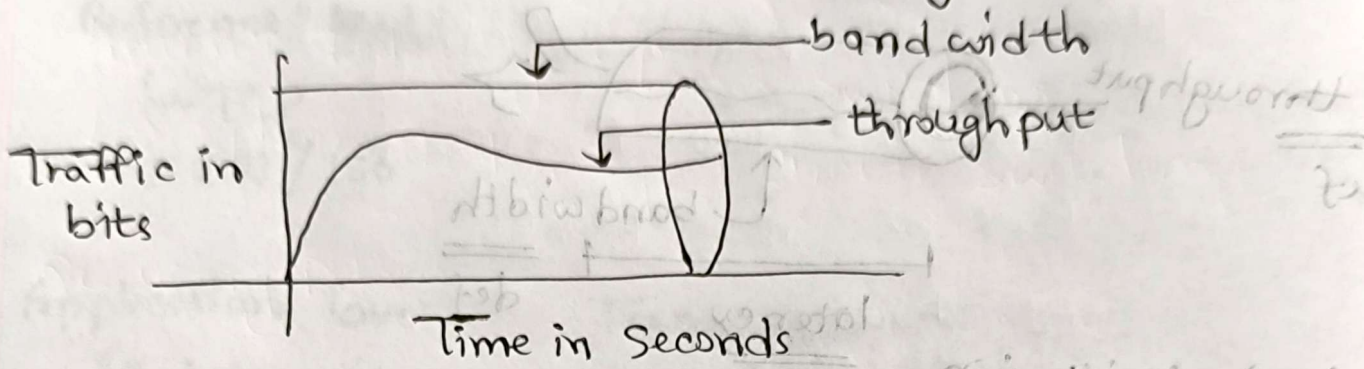
① Good put

② Bad put

data with
retransmission
and header

successful
data with
out
retransmission
data
and
header

Bandwidth vs Throughput



If there is no loss,

$\text{Bandwidth} = \text{Throughput}$ (Theoretically)

But in reality,

it is not possible

HTTP protocol

↳ status codes

Definitions

from the last slides

Summary

Throughput & Delay

RTT, Latency, Bandwidth

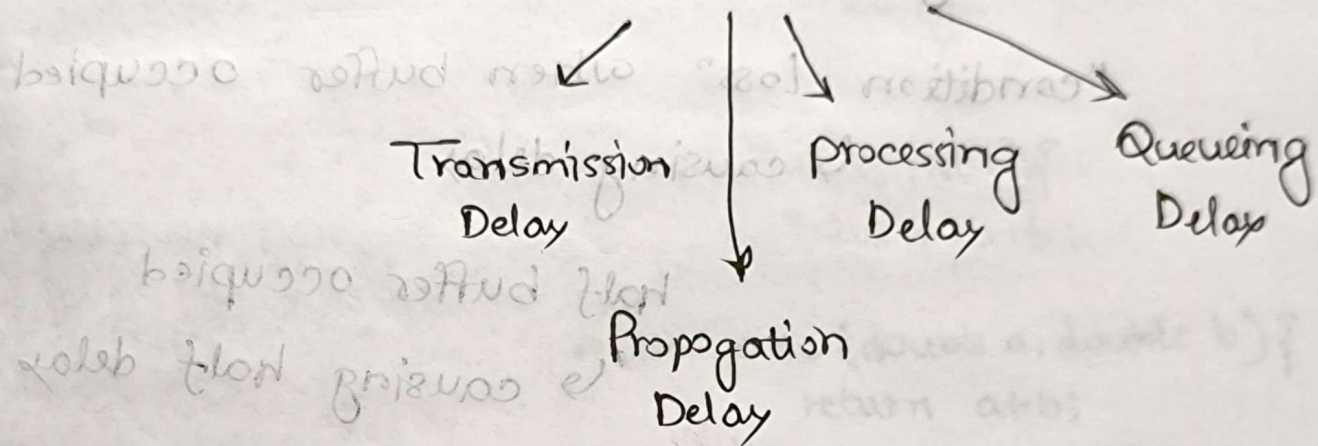
H.W: Solve the questions.

#End to End Delay

Nodal Link = One Link

Nodal Link Delay :

Nodal Link = 4 components



"Nodal Delay" from book (chapter-1. ppt)

Packet Switching : Store and Forward

Transmission Delay :

Propagation :

Packet $\rightarrow$ 1 KB (standard)

H.W: Nodal Link

Store-and-forward

E-2-E delay

Packet Switching: Queueing Delay, Loss

Internet Technology: Packet Switching

Land phone Technology: Circuit Switching

Queueing Delay:

"condition loss" when buffer occupied

↳ causing delay

half buffer occupied

↳ causing half delay

no buffer occupied

↳ no delay

Two core network functions

Forwarding

Routing